

Quiz 2: Qualitative Analysis I — The Workflow

Review of Session 2. ~10–15 minutes.

Section A: The battery

1. List the eight powder tests in the order you run them.
2. Why does running the tests in a fixed order make you faster?
3. How much sample should you use per test?

Section B: Reading results

1. Two powders both fizz with acid. One dissolves fully, one stays cloudy. Name each (with formula).
2. A powder makes an "oobleck" — liquid when stirred slowly, solid when poked fast. What is it, and what will iodine do to it?
3. A powder forms cubic crystals, dissolves fully, doesn't fizz, no iodine reaction, and crackles without melting on gentle heat. What is it?
4. What are the three parts of an ID answer that earn full points?

Section C: Dichotomous keys

1. In a key for {baking soda, chalk, cornstarch, salt, sugar, gypsum}, which single test separates the most powders at once?
2. Trace a key: a powder stays cloudy in water, does not fizz with acid, and turns iodine blue-black. What is it?

Section D: Mixtures

1. Name two signs that a "sample" is actually a mixture of two powders.
2. If a suspected mixture is really one powder, how do you know?

